

PROBLEM SET 1—SOLUTIONS

ECO403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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Instructions: Due on 11/09/19 at the beginning of the lecture. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission.

- Suppose that the set of prizes X is a finite subset of $\mathbb{R}$, and that a prize x denotes an award of x kzt. A decision-maker says that, if he knew that the true state of the world was in some set T , then we would weakly prefer a lottery f over another lottery g ($f \succeq_T g$) if and only if

$$\sum_{s \in T} \min \{x | f(x|s) > 0\} \geq \sum_{s \in T} \min \{x | g(x|s) > 0\}.$$

- Interpret the condition.
- Which of Axioms 1–8 (if any) does this preference relation violate?

Solution

Define the following function: $H : L \times \mathcal{E} \rightarrow \mathbb{R}$ as $H(f, T) = \sum_{s \in T} \min \{x | f(x|s) > 0\}$, i.e. H takes as arguments a lottery in L and an event in $\mathcal{E}$ and outputs a real number. Thus, $f \succeq_T g \Leftrightarrow H(f, T) \geq H(g, T)$.

- For any lottery f and event S , the function $H(f, S)$ considers only the *worst prizes* that the decision maker considers *possible* for any $s \in S$. The decision maker exhibits extreme pessimism about how he evaluates each lottery. 15
- We check each axiom one by one.

- Completeness and transitivity follow from the fact that for any f and T , $H(f, T) \in \mathbb{R}$ and thus when comparing between two lotteries at a certain event, we will be comparing two real numbers and the standard ordering of the real line is complete and transitive.
- Fix an event T and lotteries f and g such that $f(\cdot|t) = g(\cdot|t)$ for all $t \in T$, then the sets $\{x | f(x|t) > 0\}$ and $\{x | g(x|t) > 0\}$ are equal which implies that $H(f, T) = H(g, T)$ which shows that Relevance holds.
- Monotonicity is violated. Consider $\Omega = \{s_1, s_2\}$ and $X = \{z, x\}$ where the prizes are two real numbers such that $z > x$. Then $[z] \succ_S [x]$ for any $S \in \mathcal{E}$ but

$$\frac{2}{3}[z] + \frac{1}{3}[x] \sim_S \frac{1}{3}[z] + \frac{2}{3}[x],$$

since $H([z], S) = |S|z$, $H([x], S) = |S|x$, but $H(\frac{2}{3}[z] + \frac{1}{3}[x], S) = H(\frac{1}{3}[z] + \frac{2}{3}[x], S) = |S|x$ violating monotonicity.

- Consider any set of states Ω and any set of prizes X and suppose x, y and z are prizes in X such that $x > y > z$. Consider lotteries $[x]$, $[y]$ and $[z]$. Then for any $S \in \mathcal{E}$ we have that

$$[x] \succ_S [y] \succ_S [z],$$

since

$$H([x], S) = |S|x > H([y], S) = |S|y > H([z], S) = |S|z.$$

Notice that for any $\gamma \in (0, 1)$ we have that

$$H(\gamma[x] + (1 - \gamma)[z], S) = |S|z,$$

which implies that there does not exist a γ^* such that $[y] \sim_S \gamma^*[x] + (1 - \gamma^*)[z]$. Therefore, continuity is not satisfied.

A5. Suppose that we have two prizes such that $z_1 > z_2$, so that $[z_1] \succ_S [z_2]$ for any $S \in \mathcal{E}$. Then

$$\frac{1}{2}[z_1] + \frac{1}{2}[z_2] \sim_S [z_2],$$

which violates objective substitution.

The previous example does not work to show that this preference violates strict subjective substitution. So let me provide another one. Let $\Omega = \{s_1, s_2\}$ and $X = \{x, y\}$ where $x > y$. Consider the following lotteries

e	x	y	g	x	y	f	x	y	$\alpha e + (1 - \alpha)g$	x	y
s_1	1	0	s_1	0	1	s_1	0	1	s_1	α	$1 - \alpha$
s_2	0	1	s_2	1	0	s_2	0	1	s_2	$1 - \alpha$	α

where α is any number in the interval $(0, 1)$. Let $T = \{s_1, s_2\}$. Notice that $e \succ_T f$ and $g \succ_T f$ since $H(e, T) = H(g, T) = x + y > 2y = H(f, T)$. However $\alpha e + (1 - \alpha)g \sim_T f$ since $H(\alpha e + (1 - \alpha)g, T) = 2y$.

A6. Let S and T be two events in $\mathcal{E}$ such that $S \cap T = \emptyset$. Suppose that $f \succeq_S g$ and $f \succeq_T g$. This is equivalent to

$$\begin{aligned} H(f, S) \geq H(g, S) &\Leftrightarrow \sum_{s \in S} \min(x|f(x|s) > 0) \geq \sum_{s \in S} \min(x|g(x|s) > 0) \\ H(f, T) \geq H(g, T) &\Leftrightarrow \sum_{s \in T} \min(x|f(x|s) > 0) \geq \sum_{s \in T} \min(x|g(x|s) > 0) \end{aligned}$$

Then we have that:

$$\begin{aligned} H(f, S \cup T) &= \sum_{s \in S \cup T} \min(x|f(x|s) > 0) \\ &= \sum_{s \in S} \min(x|f(x|s) > 0) + \sum_{s \in T} \min(x|f(x|s) > 0) \\ &\geq \sum_{s \in S} \min(x|g(x|s) > 0) + \sum_{s \in T} \min(x|g(x|s) > 0) \\ &= \sum_{s \in S \cup T} \min(x|g(x|s) > 0) \\ &= H(g, S \cup T). \end{aligned}$$

Then we have that $f \succeq_{S \cup T} g$. The proof for strict subjective substitution is similar.

A7. Fix $t \in \Omega$, since X is finite, the collection $(H([x], \{t\}))_{x \in X}$ is also finite, and by axiom 1, there is a minimal and a maximal element.

A8. Fix r and t in Ω such that $f(\cdot|r) = f(\cdot|t)$ and $g(\cdot|r) = g(\cdot|t)$. By assumption, $H(f, \{r\}) \geq H(g, \{r\})$ but this is equivalent to

$$\begin{aligned} \min(x|f(x|r) > 0) &\geq \min(x|g(x|r) > 0) \\ &\Leftrightarrow \min(x|f(x|t) > 0) \geq \min(x|g(x|t) > 0) \Leftrightarrow H(f, \{t\}) \geq H(g, \{t\}). \end{aligned}$$

2. A decision-maker expresses the following preference¹ ordering for monetary lotteries (all quantities in thousands of kzt)

$$\begin{aligned} [600] &\succ [400] \succ 0.90[600] + 0.10[0] \\ &\succ 0.20[600] + 0.80[0] \succ 0.25[400] + 0.75[0] \succ [0] \end{aligned}$$

Are these preferences consistent with any state-independent utility for money? If so, show a utility function that applies. If not, show an axiom that this preference ordering violates.

¹ $\succ$ without an event subindex represents that it holds at every event.

Solution

Suppose that such a state-independent utility function existed, that is, let $U : X \rightarrow \mathbb{R}$ represents the preferences as described above. Since U is a state-independent utility that represents the preferences, then it must satisfy axioms 1–8. The preferences also will impose conditions on U that it must satisfy. We will analyze the axioms and these extra conditions in unison to reach a contradiction, implying that no U with such properties can exist.

The preferences above imply that if U represents them:

$$\begin{aligned} U(600) &> U(400) > 0.90U(600) + 0.10U(0) \\ &> 0.20U(600) + 0.80U(0) > 0.25U(400) + 0.75U(0) > U(0). \end{aligned}$$

In particular notice that $[600]$ and $[0]$ are respectively the best and worst lotteries. Thus, $[400] \sim \lambda[600] + (1 - \lambda)[0]$ where $\lambda = \frac{U(400) - U(0)}{U(600) - U(0)}$, and $0.25[400] + 0.75[0] \sim \delta[600] + (1 - \delta)[0]$ where $\delta = 0.25\lambda$; in both cases we have used the continuity and objective substitution axioms.

By monotonicity we must have that $\lambda > 0.90$ and also that $0.20 > \delta = 0.25\lambda$. The last condition implies that $0.8 > \lambda$. Therefore, we have that $0.90 < \lambda < 0.8$ which is a contradiction.

This contradiction is happening because even if we assume that all other axioms hold, the objective substitution axiom will always be violated. Consider the following: $[400] \succ 0.90[600] + 0.10[0]$. Suppose that we mix each side with the lottery $[0]$, with weights $(0.25, 0.75)$, if independence were satisfied, we would get that: $0.25[400] + 0.75[0] \succ 0.225[600] + 0.775[0]$ but this last lottery by monotonicity is strictly preferred to $0.20[600] + 0.80[0]$ which in turn is strictly preferred according to the original preferences to $0.25[400] + 0.75[0]$ which is a contradiction. Thus the objective substitution axiom fails.

3. A monopolist can produce a good in different qualities. The cost of producing a unit of quality s is s^3 . Consumers buy at most one unit and have utility function:

$$u(s|\theta) = \begin{cases} \theta s & \text{if they consume one unit of quality } s, \\ 0 & \text{if they do not consume.} \end{cases}$$

The monopolist decides on the quality (or qualities) it is going to produce and price. Consumers observe qualities and prices and decide which quality to buy if at all.

- Interpret the parameter θ of the consumers.
- Characterize the first-best solution, assuming that θ belongs to a subset $\Theta \subset \mathbb{R}$.
- How do the first-best quality depends on θ ?
- Now assume that $\Theta = \{\theta_L, \theta_H\}$ with $\theta_L < \theta_H$. Furthermore, suppose that the seller cannot observe θ but believes that $\theta = \theta_H$ with probability β and that $\theta = \theta_L$ with probability $1 - \beta$. Write the optimization program that the monopolist will need to solve.
- (Optional, attempt after Monday's lecture) Characterize the second-best solution and the consumer's optimal informational rents.

Solution

- Assume that $\theta \in \mathbb{R}_+$. Notice that $\frac{\partial u(s|\theta)}{\partial s} = \theta$ so that the marginal utility with respect to quality depends on the parameter θ . Thus, the larger the consumer's θ -parameter, the larger it is the marginal utility with respect to quality.
- For the First-Best, recall that for each type, the principal solves

$$\begin{aligned} \max_{s_\theta, P_\theta} \{ & P_\theta - s_\theta^3 \} \\ \text{s.t. } & \theta s_\theta - P_\theta \geq 0 \end{aligned} \tag{1}$$

where (s_θ, P_θ) is a type-dependent menu of qualities and transfers. Note that the participation con-

straint will bind, i.e. $P_\theta = \theta s_\theta$ and thus, the principal can solve the program:

$$\max_{q(\theta)} \{\theta s_\theta - s_\theta^3\} \quad (2)$$

The first order condition for this program is

$$\theta = 3(s_\theta^*)^2 \Rightarrow s_\theta^* = \sqrt{\frac{\theta}{3}},$$

which characterizes the optimal quantity. The previous expression together with $t_\theta^* = \theta s_\theta^*$ characterize the optimal first-best contract.

(c) For the First-Best quality

$$\frac{ds_\theta^*}{d\theta} = \frac{1}{2\sqrt{3\theta}} > 0$$

for every $\theta \in \mathbb{R}_+$. Thus quality is increasing with θ .

(d) The monopolist needs to solve the following program:

$$\begin{aligned} \max_{\{(s_L, s_H), (P_L, P_H)\}} & \{\beta(P_H - s_H^3) + (1 - \beta)(P_L - s_L^3)\} \\ \text{s.t.} & \quad \theta_H s_H - P_H \geq \theta_H s_L - P_L \\ & \quad \theta_L s_L - P_L \geq \theta_L s_H - P_H \\ & \quad \theta_H s_H - P_H \geq 0 \\ & \quad \theta_L s_L - P_L \geq 0. \end{aligned} \quad (3)$$

(e) In problem (3), the participation constraint for the low type and the incentive compatibility constraint for the high type imply the participation constraint for the high type. Furthermore, the ability of the high type to mimic the low type but not the other way around implies that the incentive compatibility constraint of the low type will be satisfied and thus can be omitted. Indeed, consider the first-best contract, which involves zero rents for both types, that contract is not incentive compatible because the θ_H buyer will prefer to choose the contract for the low type than her own first-best allocation: while this allocation restricts consumption, it nevertheless allows the high type to experience a positive informational rent. On the other hand, the low type θ_L will not find attractive to raise his consumption to the first best, since doing so would involve a transfer that, while leaving the high type with zero utility, would lead to a low type to experience a negative payoff since he has lower valuation for consumption.

Furthermore, the participation constraints for the low type and the incentive constraint for the high type will hold with equality:

$$\theta_H s_H - P_H = \theta_H s_L - P_L, \quad P_L = \theta_L s_L.$$

Solving for transfers and substituting them into problem (3) yields

$$\max_{(q_H, q_L)} \beta(\theta_H s_H - (\theta_H - \theta_L)s_L - s_H^3) + (1 - \beta)(\theta_L s_L - s_L^3). \quad (4)$$

The first order conditions for program (4) yield:

$$\begin{aligned} \theta_H - 3(s_H^{\text{SB}})^2 &= 0 \Rightarrow s_H^{\text{SB}} = \sqrt{\frac{\theta_H}{3}} \\ -\beta(\theta_H - \theta_L) + (1 - \beta)(\theta_L - 3(s_L^{\text{SB}})^2) &= 0 \Rightarrow s_L^{\text{SB}} = \max \left\{ \sqrt{\frac{\theta_L}{3} - \frac{\beta}{3(1 - \beta)}(\theta_H - \theta_L)}, 0 \right\}, \end{aligned}$$

where the last equation accounts for the fact that s_L^{SB} should be non-negative. Transfers are given by

$$\begin{aligned} t_L^{\text{SB}} &= \theta_L \max\{s_L^{\text{SB}}, 0\}, \\ t_H^{\text{SB}} &= \theta_H s_H^{\text{SB}} - (\theta_H - \theta_L)s_L^{\text{SB}}, \end{aligned}$$

The informational rent for the low type agent is zero and for the high type agent is $(\theta_H - \theta_L)s_L^{SB} \geq 0$ with strict inequality as long as s_L^{SB} is strictly positive.

Here is a graph of the solution when $\theta_H = 27$, $\theta_L = 12$ and $\beta = 1/3$. With these parameter values, we have a strictly positive solution for s_L^{SB} .

